STAC67: Regression Analysis

Lecture 16

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Multicollinearity

When the predictors are uncorrelated

Eight work crews: crew size X_1, bonus pay X_2, productivity Y. The predictors are uncorrelated below: $r(X_1, X_2) = 0$.

X_1	4	4	4	4	6	6	6	6
X_2	2	2	3	3	2	2	3	3
Y	42	39	48	51	49	53	61	60

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Model	$\hat{\beta}_1$	$\hat{\beta}_2$
$Y \sim X_1$	5.375	
$Y \sim X_2$		9.250
$Y \sim X_1 + X_2$	5.375	9.250

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The slopes do not change, and neither do the extra sums of squares: $SSR(X_1) = SSR(X_1 \mid X_2) = 231.125$ and $SSR(X_2) = SSR(X_2 \mid X_1) = 171.125$. The order no longer matters.

When the predictors are correlated

In the acceleration data, the four predictors move together:

```
       wt   hp disp  cyl
wt   1.00 0.86 0.93 0.90
hp   0.86 1.00 0.90 0.84
disp 0.93 0.90 1.00 0.95
cyl  0.90 0.84 0.95 1.00
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cyl	0.90	0.84	0.95	1.00



The coefficient of weight depends on which other predictors are in the model:

Model	$\hat{\beta}_{\text{wt}}$	$s\{\hat{\beta}_{\text{wt}}\}$
acc ~ wt	-0.00136	0.00015
acc ~ wt + hp	0.00229	0.00021
acc ~ wt + hp + disp	0.00306	0.00029
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The sign changes once horsepower is in (Lecture 10), and the standard error doubles as more correlated predictors are added.

Why the variances inflate

With two predictors, the variance of a slope is (stated without proof)

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{S_{11}\,(1 - r_{12}^2)}, \qquad S_{11} = \sum_i (X_{i1} - \bar{X}_1)^2,$$

where r_{12} is the correlation between the two predictors. With one predictor, it was σ^2 / S_{11}.

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \varepsilon \quad , r_{12} = \mathrm{corr}(X_1, X_2)$$

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The factor $1/(1 - r_{12}^2)$ is 1 when $r_{12} = 0$, and explodes as $|r_{12}| \to 1$.

Displacement and cylinders: $r = 0.951$, a factor of 10.4 on the variance:

	`acc ~ disp`	`acc ~ disp + cyl`
$s\{\hat{\beta}_{\text{disp}}\}$	0.0011	0.0036

The standard error is multiplied by about $\sqrt{10.4} \approx 3.2$.

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With more predictors, $1/(1 - R_k^2)$, where R_k^2 comes from regressing X_k on the other predictors, is the *variance inflation factor*, later in the course. For displacement among the four predictors, it is 19.3.

$$Var(\hat{\beta}_k) = \frac{\sigma^2}{S_{kk}(1 - R_k^2)}$$

Significant alone, but not once the others are in

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	20.0553	0.4061	49.3866	0
cyl	-0.8226	0.0708	-11.6248	0

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	17.8864	0.6018	29.7228	0.0000
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The t test asks whether cylinders add anything once displacement is known. With $r = 0.95$, they carry almost the same information, so the answer is: very little.

What multicollinearity does, and does not do

It does not violate any assumption: $\hat{\beta}$ is still unbiased, and, unless the dependence is exact, $\boldsymbol{X}'\boldsymbol{X}$ can still be inverted. Fitted values and predictions inside the range of the data stay valid.

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It does:

- inflate the standard errors of the coefficients;
- make the coefficients unstable: they change, and can change sign, depending on which predictors are in the model;
- blur the meaning of "the others held fixed", when the predictors rarely move separately;
- make the sequential sums of squares depend heavily on the order;
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Remedies, later in the course: drop or combine predictors, or use ridge regression.

Qualitative predictors

Coding a qualitative predictor

Many predictors are qualitative (categorical): the type of a company, a treatment group, a region. They enter the model through indicator variables (dummy variables):

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With a single binary predictor, the model is

$$Y_i = \beta_0 + \beta_1 D_i + \epsilon_i : \qquad E(Y_i) = \begin{cases} \beta_0 & \text{if } D_i = 0, \\ \beta_0 + \beta_1 & \text{if } D_i = 1. \end{cases}$$

β_0 is the mean response of the group coded 0, and β_1 is the difference between the two group means.

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Least squares gives $\hat{\beta}_0 = \bar{Y}_0$ and $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_0$, the sample means of the groups. The t test of $H_0 : \beta_1 = 0$ is the two-sample t test with a pooled variance.

Example: how fast do insurance firms innovate?

For 20 insurance firms: the number of months before the firm adopted an innovation (`months`), and whether it is a stock company ($D = 1$, 10 firms) or a mutual company ($D = 0$, 10 firms).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	16.7	2.92	5.719	0.000
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But $t = 1.31$ ($p = 0.21$): on its own, the type of firm shows no significant effect. We come back to this after adding the size of the firm.

A factor with k levels

A qualitative predictor with k levels needs $k-1$ indicators. The level without an indicator is the baseline: its mean is β_0, and each coefficient is the difference between a level and the baseline:

$$Y_i = \beta_0 + \beta_1 D_{i1} + \cdots + \beta_{k-1} D_{i,k-1} + \epsilon_i,$$

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Why not k indicators? With an intercept, the k indicators add up to the column of 1s: $D_{i1} + \cdots + D_{ik} = 1$ for every i. The columns of $\boldsymbol{X}$ are linearly dependent and $\boldsymbol{X}'\boldsymbol{X}$ cannot be inverted (Problem Set 5, Exercise 3).

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The choice of baseline does not change the fitted values; it only changes which differences the coefficients report. R takes the first level in alphabetical order.

Example: plant weights in three groups

Dried weights of 30 plants: a control group and two treatments, 10 plants each (`PlantGrowth`, in R).

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	5.032	0.197	25.527	0.000
grouptrt1	-0.371	0.279	-1.331	0.194
grouptrt2	0.494	0.279	1.772	0.088

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Baseline: `ctrl`, with mean 5.032. The coefficients are the differences $\texttt{trt1} - \texttt{ctrl} = -0.371$ and $\texttt{trt2} - \texttt{ctrl} = 0.494$.

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Are the three group means equal? $H_0 : \beta_1 = \beta_2 = 0$ is the overall F test of this model: $F^* = 4.85$ on 2 and 27 df, $p = 0.016$. We reject: the group means are not all equal (here `trt1` and `trt2` lie on either side of the control).

This is the one-way analysis of variance.

Why not code the levels $1, 2, \ldots, k$?

Back to the plants: code the groups `ctrl`, `trt1` and `trt2` as $L = 1, 2, 3$ in one column:

$$Y_i = \beta_0 + \beta_1 L_i + \epsilon_i : \qquad E(Y_i) = \beta_0 + \beta_1, \quad \beta_0 + 2\beta_1, \quad \beta_0 + 3\beta_1.$$

Going from one group to the next always adds the same amount β_1: the levels are forced to be ordered and equally spaced.

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For these data:

	`ctrl`	`trt1`	`trt2`
Sample means	5.03	4.66	5.53
Fitted means, codes $1, 2, 3$	4.83	5.07	5.32
Fitted means, codes $2, 1, 3$	5.07	4.64	5.51

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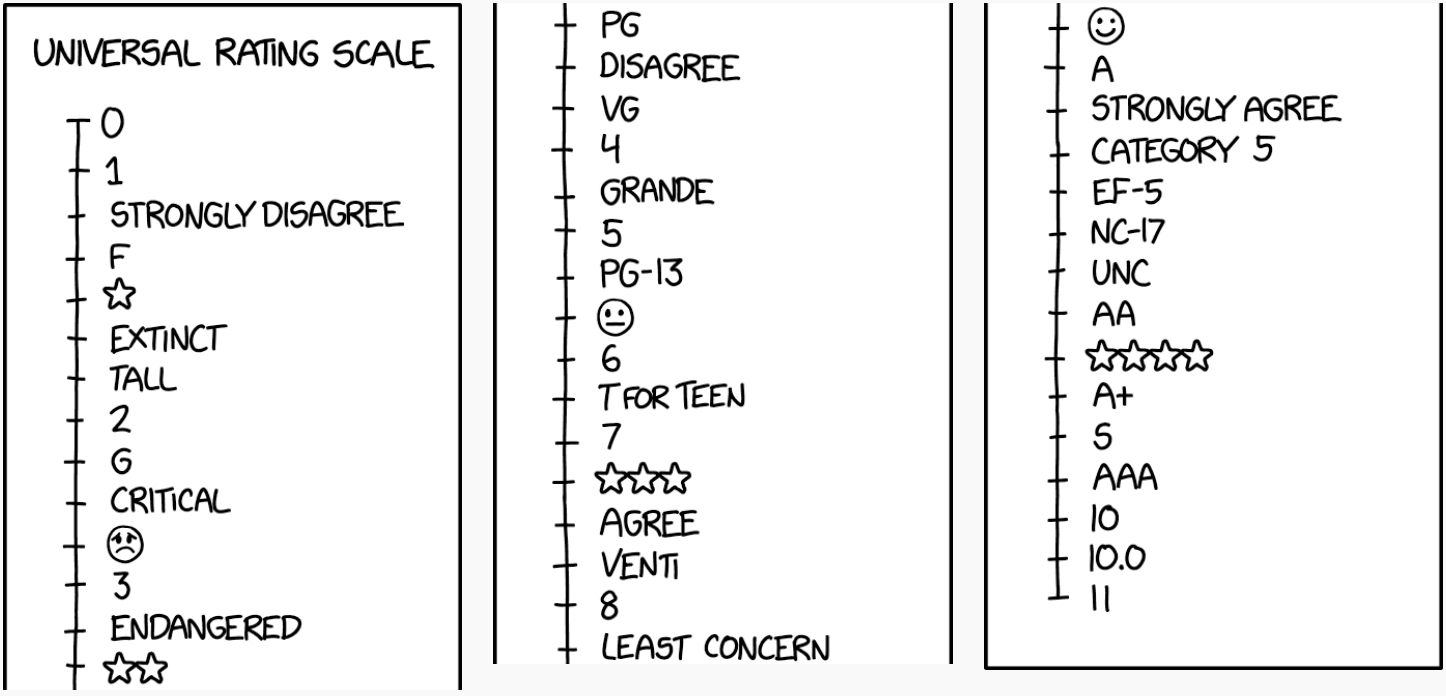
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The fit misses the means, and it changes when the labels, which are arbitrary, are swapped. With indicators, $E(Y_i) = \beta_0,\ \beta_0 + \beta_1,\ \beta_0 + \beta_2$: each level has its own mean, and the fitted means are the sample means, whatever the labels.

One scale for everything?



Coding the levels $1, 2, \ldots, k$ imposes an order, and equal gaps, that the categories do not have.

xkcd.com/2329 (CC BY-NC 2.5)

One quantitative and one qualitative predictor

Parallel lines

With a quantitative predictor X and an indicator D:

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Group	Mean response	
$D = 0$	$E(Y) = \beta_0 + \beta_1 X$	intercept β_0, slope β_1
$D = 1$	$E(Y) = (\beta_0 + \beta_2) + \beta_1 X$	intercept $\beta_0 + \beta_2$, slope β_1

Two lines with the same slope: they are parallel with a vertical distance β_2 apart.

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- β_1: change in the mean response per unit of X, in either group.
- β_2: difference between the mean responses of the two groups at the same value of X.
- $H_0 : \beta_2 = 0$ (a t test): one line for both groups.

The insurance firms, with their size

Add the size of the firm (total assets, in millions of dollars):

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	33.8741	1.8139	18.6751	0
size	-0.1017	0.0089	-11.4430	0
stock	8.0555	1.4591	5.5208	0

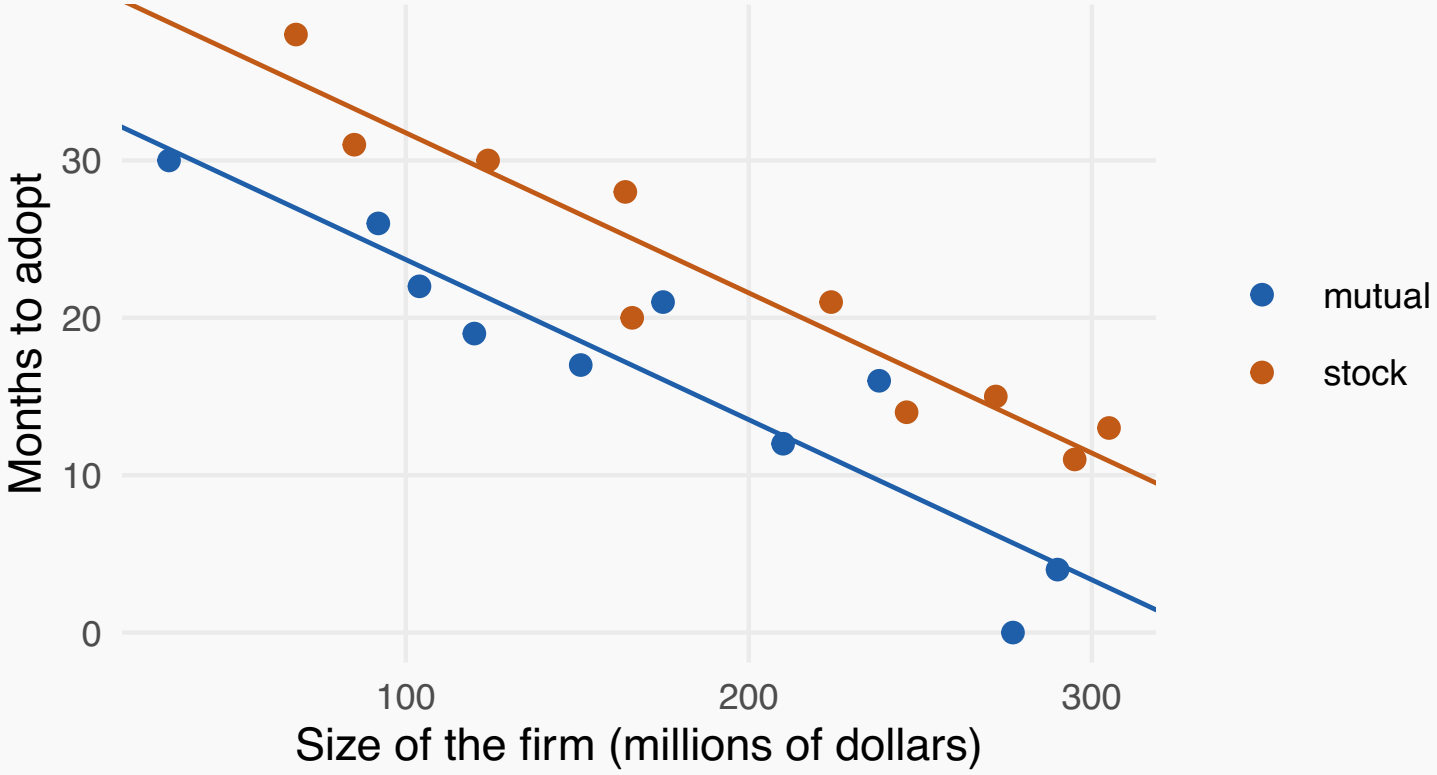
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At a given size, stock firms take about 8.1 months longer than mutual firms ($t = 5.52$, $p < 0.001$). In both groups, each extra million dollars of assets shortens the delay by 0.102 months.

The two fitted lines



Mutual: $\hat{Y} = 33.87 - 0.1017\,X$. Stock: $\hat{Y} = 41.93 - 0.1017\,X$.

Why the type of firm mattered only once size was in

Model	$\hat{\beta}$ for `stock`	$s\{\hat{\beta}\}$	$\sqrt{MSE}$
`months` $\sim$ `stock`	5.40	4.13	9.23
`months` $\sim$ `size + stock`	8.06	1.46	3.22

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- Size explains most of the variation in the delay, so $\sqrt{MSE}$ drops from 9.2 to 3.2 months, and the standard error with it.
- Stock firms in the sample are larger (195 against 169 million on average), and larger firms innovate sooner. Comparing the raw means mixes the two effects; the model compares firms of the same size.

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The same idea as "holding the other predictors fixed" in Lecture 10, with a qualitative predictor.

With k groups

With a factor of k levels and one quantitative predictor:

$$Y_i = \beta_0 + \beta_1 X_i + \gamma_2 D_{i2} + \cdots + \gamma_k D_{ik} + \epsilon_i :$$

k parallel lines, one per level, with intercepts β_0 (baseline) and $\beta_0 + \gamma_j$.

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"Do the groups differ, at a given X?" is $H_0 : \gamma_2 = \cdots = \gamma_k = 0$: $k - 1$ restrictions, tested with full and reduced models, as for the extra sums of squares:

$$F^* = \frac{\big(SSE(X) - SSE(X, D_2, \ldots, D_k)\big)/(k-1)}{MSE(X, D_2, \ldots, D_k)},$$

which follows an $F(k-1,\ n-k-1)$ distribution under H_0.

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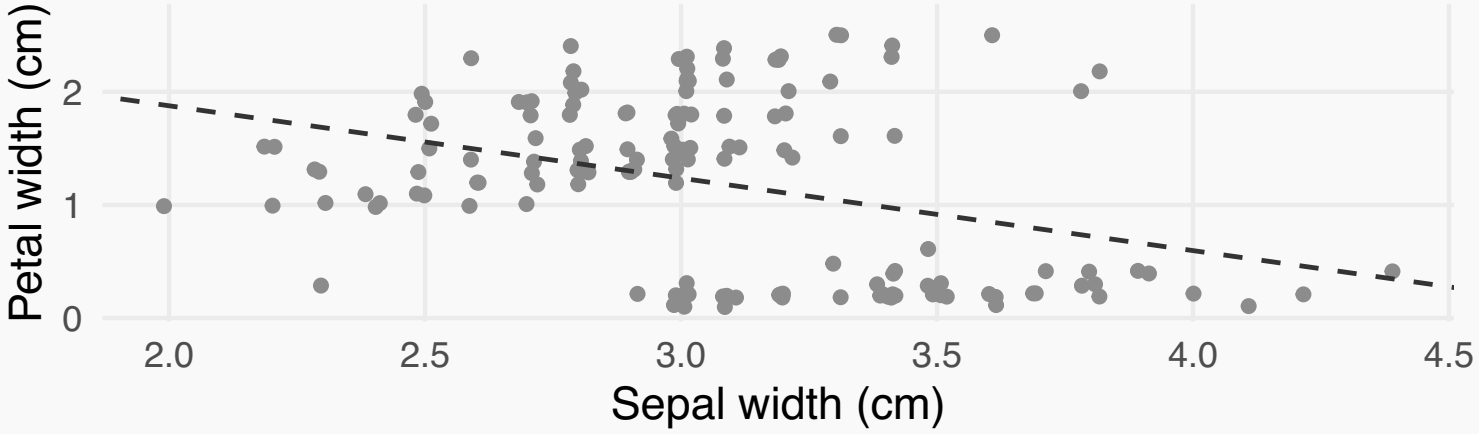
$$F^* = \frac{\big(SSE(X) - SSE(X, D_2, \ldots, D_k)\big)/(k-1)}{MSE(X, D_2, \ldots, D_k)},$$

which follows an $F(k-1,\ n-k-1)$ distribution under H_0.

In R: `anova(lm(y ~ x), lm(y ~ x + group))`, with `group` a factor.

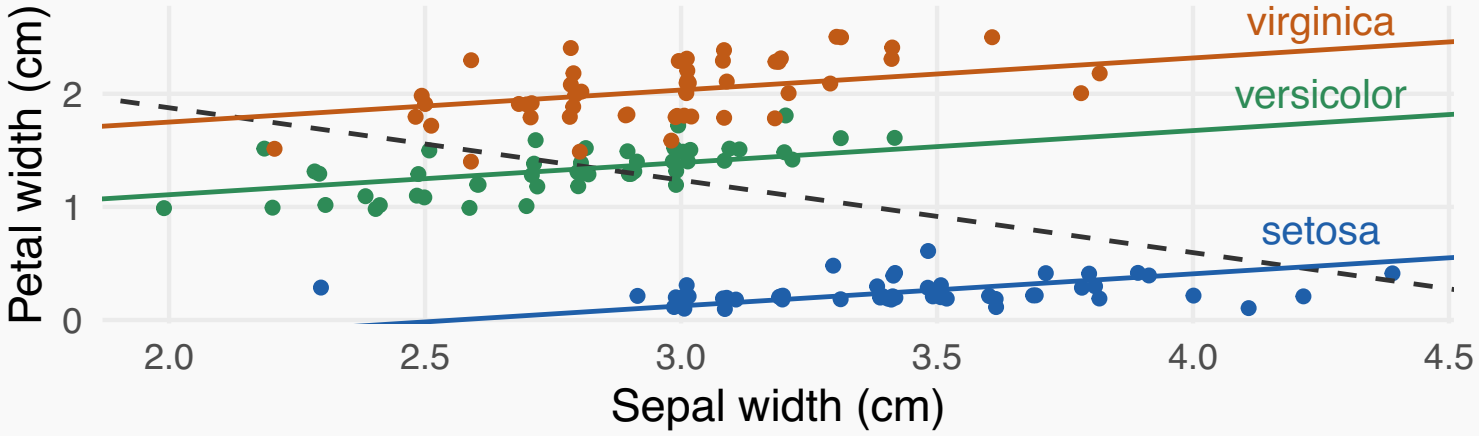
Simpson's paradox

150 iris flowers, 50 from each of three species (`iris`, in R):



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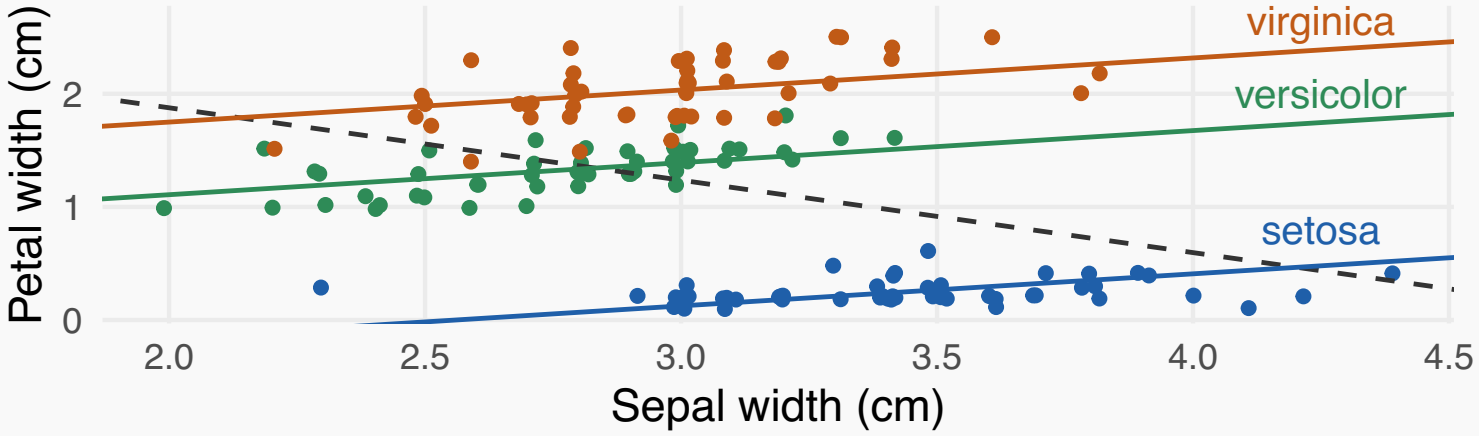
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This is Simpson's paradox: an association that reverses once a third variable is taken into account. Setosa flowers have the widest sepals but by far the narrowest petals, so the dashed line mixes the comparison within a species with the differences between species.

Interactions

When the slopes may differ

The parallel-lines model assumes that X has the same effect in both groups: the model is additive. An interaction term, the product $D \cdot X$, relaxes this:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 D_i + \beta_3 D_i X_i + \epsilon_i.$$

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- β_3: how much the slope of X changes from group 0 to group 1.
- β_2: the difference between the groups at $X = 0$ only. With an interaction, the difference between the groups depends on X: it is $\beta_2 + \beta_3 X$.

Two tests

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In R, `y ~ x * d` fits x, d and their product $x{:}d$; `anova(lm(y ~ x), lm(y ~ x * d))` gives the test of the same line.

Insurance firms: is the slope the same?

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	33.8384	2.4406	13.8645	0.0000
size	-0.1015	0.0131	-7.7786	0.0000
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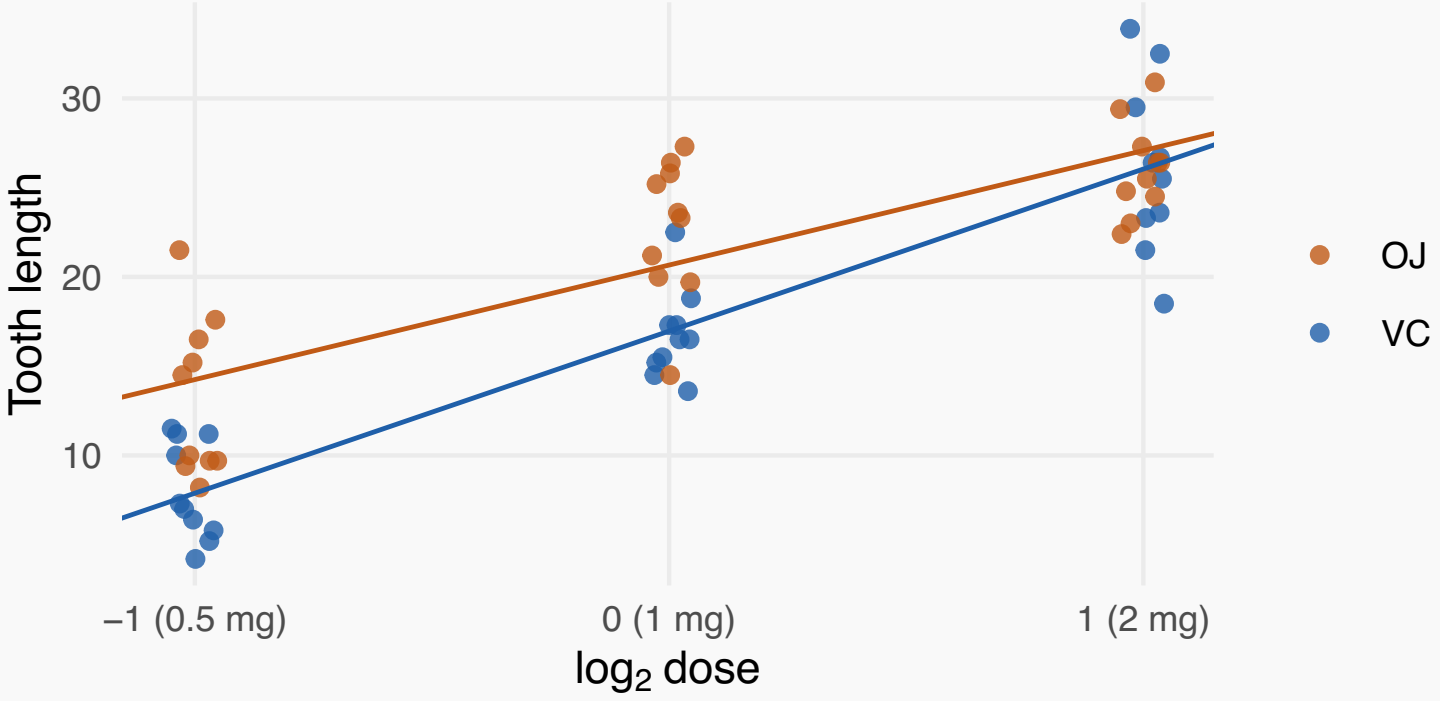
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Same line? $F^* = 14.34$ on 2 and 16 df ($p = 0.0003$): one line for both types is rejected; the intercepts differ.

Example: vitamin C and tooth growth

Tooth growth of 60 guinea pigs (`ToothGrowth`, in R): vitamin C at a dose of 0.5, 1 or 2 mg per day, given as orange juice (OJ) or as ascorbic acid (VC). We use $X = \log_2(\text{dose}) = -1, 0, 1$.



The lines of the interaction model: ascorbic acid starts lower but rises faster.

The interaction model

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	20.663	0.679	30.425	0.000
ldose	6.415	0.832	7.712	0.000
suppVC	-3.700	0.960	-3.852	0.000
ldose:suppVC	2.665	1.176	2.266	0.027

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- Orange juice (baseline): slope 6.42 per doubling of the dose. Ascorbic acid: 6.42 + 2.66 = 9.08.
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Difference VC $-$ OJ: $-3.70 + 2.665\,X$: -6.36 at 0.5 mg, -3.70 at 1 mg, -1.04 at 2 mg.
The coefficient of `suppVC` alone is the difference at 1 mg only.

Interaction between two qualitative predictors

Tooth growth with the dose as a factor (3 levels) and the supplement (2 levels): 6 cells. Mean tooth length in each cell:

	0.5 mg	1 mg	2 mg
OJ	13.23	22.70	26.06
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Test of the interaction: the 2 extra parameters, $F^* = 4.11$ on 2 and 54 df ($p = 0.022$): the supplement matters less at the highest dose.

Three rules when using interactions

- Keep the main effects. When $D \cdot X$ is in the model, keep D and X too, even if their own tests are not significant: without them, the two lines are forced through constraints that rarely make sense.

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- Count the parameters. Each interaction with a factor of k levels adds $k - 1$ parameters; with several predictors, they add up quickly (and $(X'X)^{-1}$ may not exist anymore).

Exercise

For the insurance firms, `lm(months ~ size * stock)` gives, with X the size of the firm and $D = 1$ for stock firms:

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size:stock   -0.0004     0.0183 -0.0228   0.9821
```

1. Write down the fitted line of each type of firm.
2. Estimate the difference in mean months between a stock and a mutual firm of size 200.
3. Which model would you report, this one or the parallel-lines model, and why?

Exercise: solution

1. Mutual ($D = 0$): $\hat{Y} = 33.84 - 0.1015\, X$. Stock ($D = 1$): $\hat{Y} = 41.97 - 0.1019\, X$.

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3. The parallel-lines model. In the table, the test of $H_0 : \beta_3 = 0$ (same slope) gives $t = -0.02$, $p = 0.98$: no evidence that the slopes differ, and they are almost equal (-0.1015 and -0.1019). The simpler model is easier to interpret: one slope, and one difference between the types, 8.1 months ($t = 5.52$), that holds at every size.

Summary

- Correlated predictors inflate the variances of the coefficients and make them hard to interpret, without violating any assumption.
- A qualitative predictor with k levels enters through $k - 1$ indicators, and each coefficient compares a level with the baseline. Testing that all the group means are equal is the one-way analysis of variance (global F-test).
- With a quantitative predictor and an indicator: parallel lines, and the coefficient of the indicator compares the groups at the same X.
- An interaction lets the effect of one predictor depend on another: $D \cdot X$ (different slopes), or two factors (cell means).
- Same slope: t test on the interaction. Same line: F test of full against reduced model.

Next week: comparing regression functions across groups, interactions between two quantitative predictors, then polynomial regression and transformations.